

Direction of vorticity and continuation for 3D Navier–Stokes: the $\frac{1}{2}$ -Hölder criterion on the maximal classical interval

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31 August 2026

Abstract

Let u be the unique classical solution of the 3D incompressible Navier–Stokes equations on the maximal interval $[0, T^*)$, viscosity $\nu > 0$. Write $\omega = \nabla \times u$, $r = |\omega|$, $\xi = \omega/r$ on $\{r > 0\}$, and $S = \frac{1}{2}(\nabla u + (\nabla u)^\top)$. If there exist $\Omega = \Omega(u_0, \nu) > 0$ and $C = C(u_0, \nu) < \infty$ such that

$$|\xi(x, t) - \xi(y, t)| \leq C |x - y|^{1/2}$$

whenever $r(x, t), r(y, t) \geq \Omega$ and $t < T^*$, then $T^* = \infty$ and u is globally smooth. The direction ξ is never evaluated on $\{r = 0\}$. No additional geometric term is added to the PDE. The implication is the Beirão da Veiga–Berselli $\frac{1}{2}$ -Hölder form of the Constantin–Fefferman criterion. The Clay problem remains open unless Ω and C are produced from (u_0, ν) alone.

MSC-style keywords: Navier–Stokes, vorticity direction, Beale–Kato–Majda, Constantin–Fefferman, Beirão da Veiga–Berselli.

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No tether term is used. The implication proved here is the geometric regularity criterion of Constantin–Fefferman and Beirão da Veiga–Berselli, recast on $[0, T^)$ with degeneracy of $\xi = \omega/|\omega|$ respected. It does not construct the Hölder modulus from the PDE, and it does not solve the Clay problem.

1 Scope

This note treats the unmodified Cauchy problem for 3D incompressible Navier–Stokes, written as (1)–(2) below, on the maximal classical existence interval $[0, T^*)$. The only geometric input is a uniform spatial $\frac{1}{2}$ -Hölder bound on the vorticity *direction* on the set where the vorticity *magnitude* is at least a positive threshold. From that input one concludes $T^* = \infty$.

The following are **not** proved: that every smooth divergence-free u_0 produces such an Ω and C ; that $T^* < \infty$ is impossible without extra hypotheses; that a named tether or metriplectic correction is present in the equations. The PDE used is (1).

What *would* finish the Clay problem, given Theorem A, is a proof that $\Omega(u_0, \nu)$ and $C(u_0, \nu)$ exist for every smooth divergence-free u_0 . That is a different theorem.

2 The equations

Let $\nu > 0$. On $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ (the whole-space case is identical up to replacing the periodized Newtonian potential by $1/(4\pi|x|)$) one considers

$$\partial_t u + (u \cdot \nabla)u = \nu \Delta u - \nabla p, \quad (1)$$

$$\nabla \cdot u = 0, \quad (2)$$

with $u(\cdot, 0) = u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$, $\nabla \cdot u_0 = 0$. Local well-posedness in H^s , $s > 5/2$, yields a unique classical solution on a maximal interval $[0, T^*)$, $T^* \in (0, \infty]$, with the standard blow-up alternative: if $T^* < \infty$ then

$$\limsup_{t \uparrow T^*} \|\nabla u(t)\|_{L^2(\mathbb{T}^3)} = \infty. \quad (3)$$

Equivalently, $\limsup_{t \uparrow T^*} \|\omega(t)\|_{L^2} = \infty$, since $\|\omega\|_{L^2} = \|\nabla u\|_{L^2}$ under (2). This is the energy-class continuation criterion (Kato); it is the only continuation tool used below. (Beale–Kato–Majda in $L_t^1 L_x^\infty$ is stronger than needed once enstrophy is bounded.)

Curl of (1), using (2), gives the vorticity equation

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega, \quad \omega = \nabla \times u. \quad (4)$$

Write $r := |\omega|$ and, on the open set $\{r > 0\}$,

$$\xi := \frac{\omega}{r}. \quad (5)$$

The strain is the symmetric gradient

$$S := \frac{1}{2}(\nabla u + (\nabla u)^\top). \quad (6)$$

On $\{r > 0\}$ one has the stretching identity

$$(\omega \cdot \nabla)u \cdot \omega = r^2 (S\xi \cdot \xi). \quad (7)$$

The left-hand side of (7) extends continuously by 0 at $r = 0$. The unit field ξ is *not* extended; every estimate below that mentions $\xi(x)$ is restricted to $r(x) > 0$.

3 The criterion

Theorem A (Beirão da Veiga–Berselli, recast). *Let u be the classical solution of (1)–(2) on $[0, T^*)$. Suppose there exist $\Omega = \Omega(u_0, \nu) > 0$ and $C = C(u_0, \nu) < \infty$ such that, for all $t \in [0, T^*)$ and all $x, y \in \mathbb{T}^3$ with $r(x, t) \geq \Omega$ and $r(y, t) \geq \Omega$,*

$$|\xi(x, t) - \xi(y, t)| \leq C |x - y|^{1/2}. \quad (8)$$

Then $T^ = \infty$, and $u \in C^\infty(\mathbb{T}^3 \times [0, \infty))$.*

The Hölder condition may be localized to $|x - y| < \delta$ for an arbitrary $\delta > 0$; the far field is controlled by the integrable periodized kernel. Only pairs in which *both* magnitudes meet the threshold Ω are constrained. Pairs with $r(x) < \Omega$ or $r(y) < \Omega$ are estimated crudely, without using ξ .

Remark 1. On $\{r(x), r(y) > 0\}$ one has $|\xi(x) - \xi(y)| = 2|\sin(\theta/2)|$ and $\sin \theta \leq |\xi(x) - \xi(y)|$, where $\theta = \angle(\omega(x), \omega(y))$. Thus (8) implies $\sin \theta(x, y, t) \leq C|x - y|^{1/2}$ on the high-high set, which is Assumption A of Beirão da Veiga–Berselli [1] at exponent $\alpha = 1/2$ with g a constant. Lipschitz direction (Constantin–Fefferman [3]) is the case $\alpha = 1$.

4 Enstrophy and stretching

Take the L^2 inner product of (4) with ω . After an integration by parts (justified on every compact subinterval of $[0, T^*)$ by classical smoothness),

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 = \int_{\mathbb{T}^3} \mathcal{K}(x, t) dx, \quad (9)$$

where the stretching density is

$$\mathcal{K}(x, t) := ((\omega \cdot \nabla) u \cdot \omega)(x, t). \quad (10)$$

On $\{r = 0\}$ one has $\mathcal{K} = 0$. On $\{r > 0\}$ one has $\mathcal{K} = r^2(S\xi \cdot \xi)$.

Biot–Savart on $\mathbb{T}^3$ reconstructs u from ω by the periodized Newtonian potential G . Differentiating under the principal value gives S as a Calderón–Zygmund operator of ω . The Constantin–Fefferman representation [3] then yields the pointwise bound, for a dimensional constant c_0 independent of u ,

$$|\mathcal{K}(x)| \leq c_0 \int_{\mathbb{T}^3} \frac{r(x)^2 r(y) \sin \theta(x, y)}{|x - y|^3} dy, \quad (11)$$

where the integrand is read as 0 if $r(x) = 0$ or $r(y) = 0$, so that ξ is never invoked on the degeneracy set. (On the torus, $|x - y|$ denotes chordal/periodized distance; the local singularity is the same as in $\mathbb{R}^3$.)

5 High-high versus complementary pairs

Fix $\Omega > 0$ as in Theorem A. Split the integral in (11) according to whether $r(y) \geq \Omega$ or not. Write $\mathcal{K} = \mathcal{K}_{\text{hh}} + \mathcal{K}_{\text{c}}$.

High-high. If $r(x) \geq \Omega$ and $r(y) \geq \Omega$, (8) and $\sin \theta \leq |\xi(x) - \xi(y)|$ give $\sin \theta / |x - y|^3 \leq C|x - y|^{-5/2}$. Hence

$$|\mathcal{K}_{\text{hh}}(x)| \leq c_0 C r(x)^2 I(x), \quad I(x) := \int_{\mathbb{T}^3} \frac{r(y)}{|x - y|^{5/2}} dy. \quad (12)$$

The field I is a Riesz potential of order $\beta = 1/2$ applied to r . Hardy–Littlewood–Sobolev on $\mathbb{T}^3$ (or $\mathbb{R}^3$) yields

$$\|I\|_{L^3} \leq C_{\text{HLS}} \|\omega\|_{L^2}, \quad (13)$$

because $1/3 = 1/2 - (1/2)/3$. If $r(x) < \Omega$, the factor $r(x)^2$ already places $\mathcal{K}_{\text{hh}}(x)$ in the complementary class treated next; alternatively one may restrict (12) to $\{r(x) \geq \Omega\}$ and estimate the remainder with $\sin \theta \leq 1$ and $r(x) < \Omega$.

Complementary pairs. If $r(y) < \Omega$, use $\sin \theta \leq 1$ and do *not* form $\xi(y)$. After the usual cancellation in the CZ kernel of S (mean-zero homogeneous kernel of degree -3), the complementary stretching is controlled by the Hardy–Littlewood maximal function of the truncated vorticity:

$$|\mathcal{K}_c(x)| \leq c_1 r(x)^2 M(r 1_{\{r < \Omega\}})(x). \quad (14)$$

In particular, using $r 1_{\{r < \Omega\}} \leq \Omega$ and the boundedness of $M : L^2 \rightarrow L^2$,

$$\int_{\mathbb{T}^3} |\mathcal{K}_c| \, dx \leq c_2 \Omega \|\omega\|_{L^2} \|\omega\|_{L^6}. \quad (15)$$

(The same bound follows from $|S(1_{\{r < \Omega\}}\omega)| \lesssim M(r 1_{\{r < \Omega\}})$. No division by r occurs.)

6 Closing the enstrophy inequality

Combining (9), (12), (13) and (15),

$$\int |\mathcal{K}_{hh}| \, dx \leq c_0 C \int r^2 I \, dx \leq c_0 C \|\omega\|_{L^3}^2 \|I\|_{L^3} \leq c_3 C \|\omega\|_{L^2} \|\omega\|_{L^6} \|\omega\|_{L^2},$$

where the interpolation $\|\omega\|_{L^3}^2 \leq \|\omega\|_{L^2} \|\omega\|_{L^6}$ was used. Thus

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 \leq c_4 (C + \Omega) \|\omega\|_{L^2}^2 \|\omega\|_{L^6}. \quad (16)$$

Sobolev on $\mathbb{T}^3$ gives $\|\omega\|_{L^6} \leq C_S \|\nabla \omega\|_{L^2}$. Young's inequality, for every $\varepsilon > 0$,

$$\|\omega\|_{L^2}^2 \|\omega\|_{L^6} \leq \frac{\varepsilon}{c_4(C + \Omega)} \|\omega\|_{L^6}^2 + \frac{c_4(C + \Omega)}{4\varepsilon} \|\omega\|_{L^2}^4,$$

so choosing ε small enough to absorb the first term into $\nu \|\nabla \omega\|_{L^2}^2$ produces

$$\frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 \leq K(C, \Omega, \nu) \|\omega\|_{L^2}^4, \quad (17)$$

with K finite and depending only on (C, Ω, ν) and on absolute Sobolev/HLS constants of $\mathbb{T}^3$.

7 Continuation

Energy equality for (1) on $[0, T^*)$ yields

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 \, ds = \frac{1}{2} \|u_0\|_{L^2}^2, \quad (18)$$

hence $\int_0^t \|\omega(s)\|_{L^2}^2 \, ds \leq (2\nu)^{-1} \|u_0\|_{L^2}^2$ for every $t < T^*$. Drop the dissipation in (17) and apply Gronwall:

$$\|\omega(t)\|_{L^2}^2 \leq \|\omega(0)\|_{L^2}^2 \exp\left(K \int_0^t \|\omega(s)\|_{L^2}^2 \, ds\right) \leq \|\omega(0)\|_{L^2}^2 \exp\left(K(2\nu)^{-1} \|u_0\|_{L^2}^2\right). \quad (19)$$

The right-hand side is independent of $t < T^*$. Therefore $\sup_{t < T^*} \|\nabla u(t)\|_{L^2} < \infty$. The blow-up alternative (3) forces $T^* = \infty$.

Once the solution exists globally in H^1 with uniformly bounded enstrophy, standard parabolic bootstrapping for (1) (Kato, or L^p - L^q estimates for the Stokes semigroup) upgrades u to $C^\infty(\mathbb{T}^3 \times [0, \infty))$.

This is Theorem A.

8 What would finish Clay

Theorem A is a *conditional* regularity theorem. Its hypotheses Ω and C are not constructed. For the Clay problem as stated by Fefferman [4] one needs, for *every* smooth divergence-free u_0 , a globally smooth solution of (1)–(2). By Theorem A it would suffice to prove:

For every smooth divergence-free u_0 and every $\nu > 0$, there exist $\Omega = \Omega(u_0, \nu) > 0$ and $C = C(u_0, \nu) < \infty$ such that (8) holds on the maximal classical interval.

That statement is not proved here. It is also not implied by local smoothness on $[0, T^*)$: a classical solution may a priori lose Hölder control of ξ on $\{r \geq \Omega\}$ as $t \uparrow T^*$ even while remaining smooth on each compact subinterval. Producing Ω, C from (u_0, ν) is the outstanding geometric problem isolated by Constantin–Fefferman and by Beirão da Veiga–Berselli. Values $\beta < 1/2$ are not known to suffice [2].

No term was added to (1). Degeneracy of ξ on $\{r = 0\}$ was respected throughout.

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